EXPERIMENT 1

PROBLEM STATEMENT - Installation and set-up of Natural Language Tool Kit (NLTK).

THEORY -

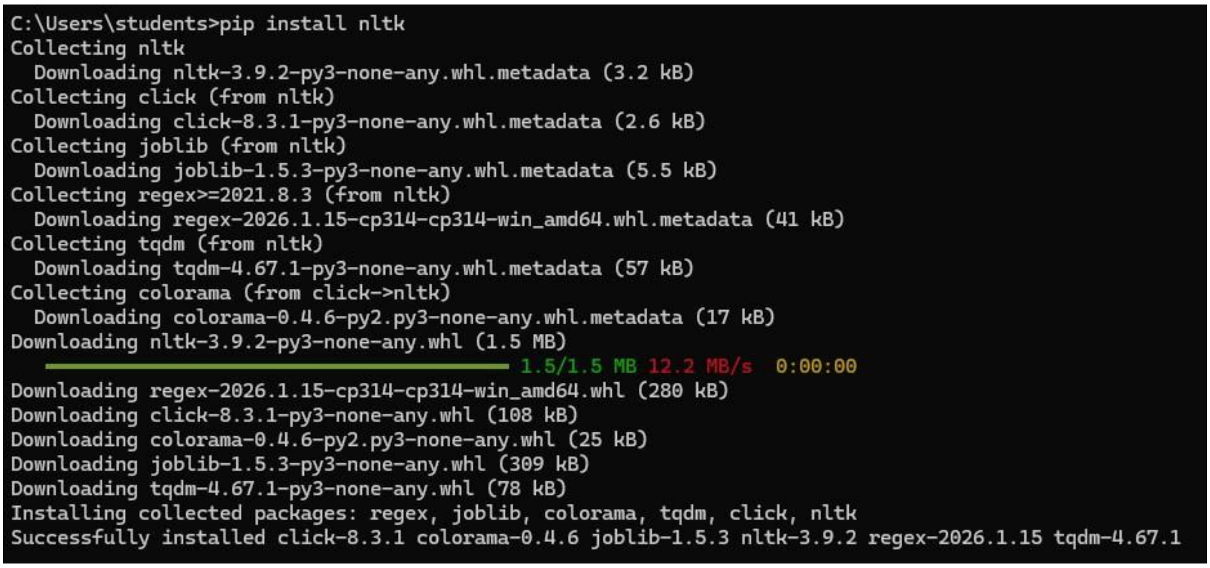
NLTK (Natural Language Toolkit) is a comprehensive Python library for natural language processing (NLP), providing essential tools for text analysis including tokenization, stemming, part-of-speech tagging, parsing, and semantic analysis, along with pre-built corpora and linguistic datasets for research and development. It is widely used in academic and industry applications for computational linguistics research and practical NLP implementations.

NLTK enables computational analysis of human language data through modular components and well-designed architectures. The toolkit is built on object-oriented principles, making it extensible and easy to integrate into larger applications. It supports various NLP tasks from basic text processing to advanced machine learning techniques. The library includes statistical models trained on diverse corpora, enabling accurate language processing. NLTK's design philosophy emphasizes educational value and research accessibility, making it ideal for students and professionals learning NLP fundamentals.

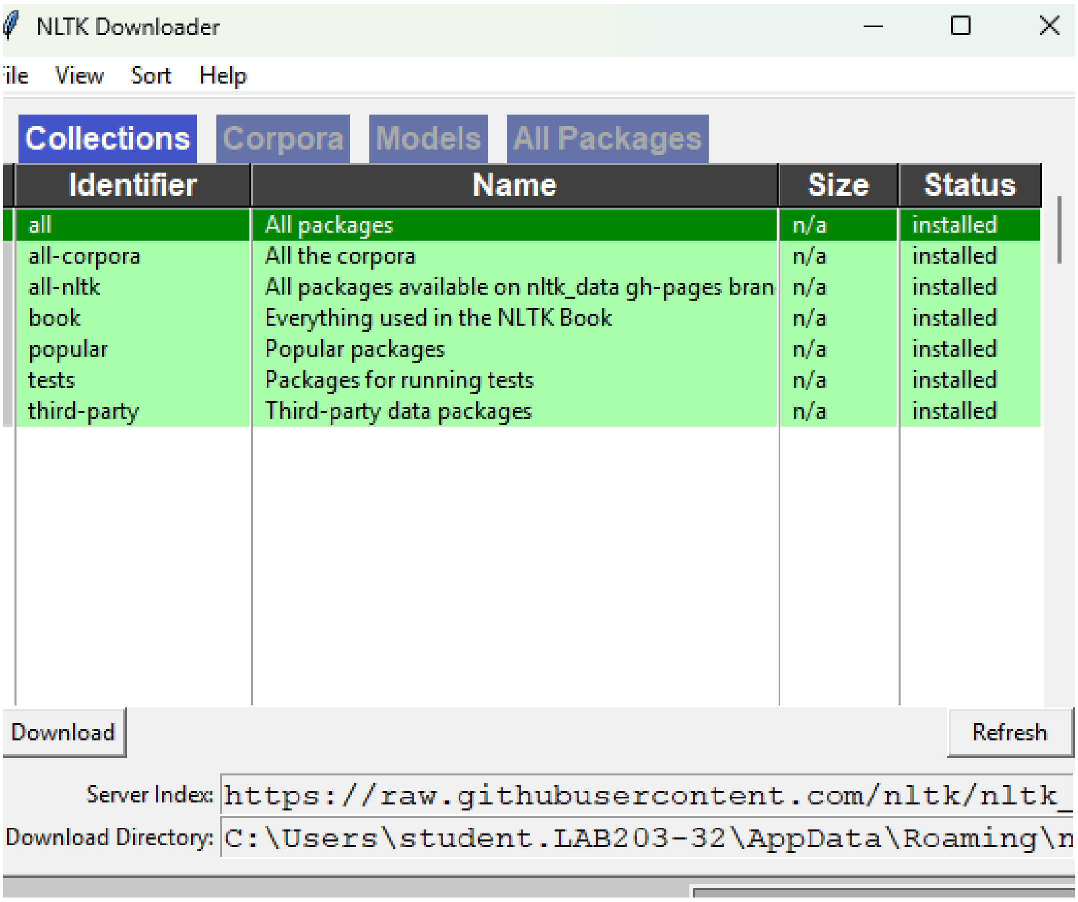
NLTK also plays a crucial role in preprocessing text data, which is a vital step in many artificial intelligence and machine learning systems. Tasks such as text normalization, stop-word removal, and lemmatization can be efficiently performed using NLTK, improving the quality of input data for further analysis. Its interactive documentation and tutorials help users understand NLP concepts through hands-on practice.Overall, NLTK serves as a foundational platform for learning and implementing natural language processing techniques. Its flexibility, rich feature set, and strong community support make it an essential toolkit for developing intelligent language-based applications such as chatbots, sentiment analysis systems, information retrieval systems, and text classification models.

CODE -

Step 1 : Install NLTK library using command prompt This step installs the NLTK package in the Python environment using the pip package manager.



```
C:\Users\students>pip install nltk
Collecting nltk
  Downloading nltk-3.9.2-py3-none-any.whl.metadata (3.2 kB)
Collecting click (from nltk)
  Downloading click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting joblib (from nltk)
  Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)
Collecting regex>=2021.8.3 (from nltk)
  Downloading regex-2026.1.15-cp314-cp314-win_amd64.whl.metadata (41 kB)
Collecting tqdm (from nltk)
  Downloading tqdm-4.67.1-py3-none-any.whl.metadata (57 kB)
Collecting colorama (from click->nltk)
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Downloading nltk-3.9.2-py3-none-any.whl (1.5 MB)
   ---------------------------------------- 1.5/1.5 MB 12.2 MB/s  0:00:00
Downloading regex-2026.1.15-cp314-cp314-win_amd64.whl (280 kB)
Downloading click-8.3.1-py3-none-any.whl (108 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading joblib-1.5.3-py3-none-any.whl (309 kB)
Downloading tqdm-4.67.1-py3-none-any.whl (78 kB)
Installing collected packages: regex, joblib, colorama, tqdm, click, nltk
Successfully installed click-8.3.1 colorama-0.4.6 joblib-1.5.3 nltk-3.9.2 regex-2026.1.15 tqdm-4.67.1
```



Step 2: Install NLTK library

The NLTK library is imported into Python to make its natural language processing modules available.

```
!pip install nltk
```

Step 3: Check the version of NLTK library

The installed version of NLTK is verified to ensure proper installation and compatibility.

```
import nltk
nltk.__version__
```

Step 4: Download the functions of NLTK library

Required NLTK datasets and resources are downloaded to enable text processing operations.

```
nltk.download('all')
```

Step 5: Print the directory to check for all the functions of NLTK library

This step displays the available NLTK modules and functions for exploration and usage.

```
print(dir(nltk))
```

Step 6: Implement word_tokenize function

The word_tokenize function is used to divide input text into individual word token

```
from nltk.tokenize import word_tokenize
text= "I am learning Natural Language Processing."
tokens= word_tokenize(text)
print(tokens)
```

OUTPUT -

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
```

```
import nltk
nltk.__version__
```

```
'3.9.1'
```

<u>LEARNING OUTCOME</u> - Through this program, I understood the fundamentals of Natural Language Processing and the role of nltk library in text analysis and was able to implement functions like tokenization.